

CHAPTER 2

Windows System and Network Administration

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Windows Network concepts

- Microsoft Windows LAN is configured using one of these two models:
 - **Workgroup**
 - **Domain**
- The model determines how users are organized.

Workgroups

- In computer networking, a workgroup is a collection of computers on a local area network (LAN) that **share common resources and responsibilities**.
- The term is most commonly associated with Microsoft Windows workgroups but also applies to other environments.
- Windows workgroups can be found in **homes, schools and small businesses**.

Workgroups Cont.

- Treats each computer in the network as an equal, or peer
 - Also called **peer-to-peer networking**
- **Each computer is a client and a server**
 - When you allow others to access resources on your computer, your computer is acting as a server
 - When you access resources on another computer, your computer is acting as a client
- Appropriate for networks with **10 or less computers**

Workgroups Cont.

- **Disadvantages of Workgroup:**
 - Most users do not want to administer resources on their computer.
 - Need user names and passwords of users who need resources.
 - Difficult to keep track of changing passwords.

Server Domain

- **Windows domains** support **client-server** local networks.
- A specially configured computer called the **Domain Controller** running a Windows Server operating system serves as a central server for all clients.
- Windows domains can handle **much more computers** than workgroups due to maintaining **centralized resource sharing and access control**.
- A client PC can belong **only to a workgroup or to a Windows domain** but not both - assigning a computer to the domain automatically removes it from the workgroup.

Server Domain Cont...

- One or more servers centralized control
- **Computers are part of a domain**
 - Single, centralized logon
 - Single point of control
 - Users can be given access to **resources anywhere in the domain**

Domain Controller

- A **domain controller** is a server that responds to authentication requests and verifies users on computer networks.
- Domains are a **hierarchical** way of organizing users and computers that work together on the same network. The domain controller keeps all of that data organized and secured.

Domain Controller Cont.

- The **primary responsibility** of the DC is to **authenticate** and **validate** user access on the network.
- When users log into their domain, the DC checks their **username, password, and other credentials** to either allow or deny access for that user.
- Domain controllers contain the data that determines and validates access to your network, including any group policies and all computer names.

Benefits and Limitation of Domain Controller

Benefits

- Centralized user management
- Enable resource sharing for files and printers.
- Avoid redundancy.
- Distributed and replicated across large network.
- Provide encryption

Limitation

- Target for cyber attack.
- Network is dependent of Domain controller uptime.
- OS should be maintained to be stable, secure and up-to-

Directory Services **Active** **Directory**

- Three main parts
 - **Domain**
 - **Tree**
 - **Forest**

Domains

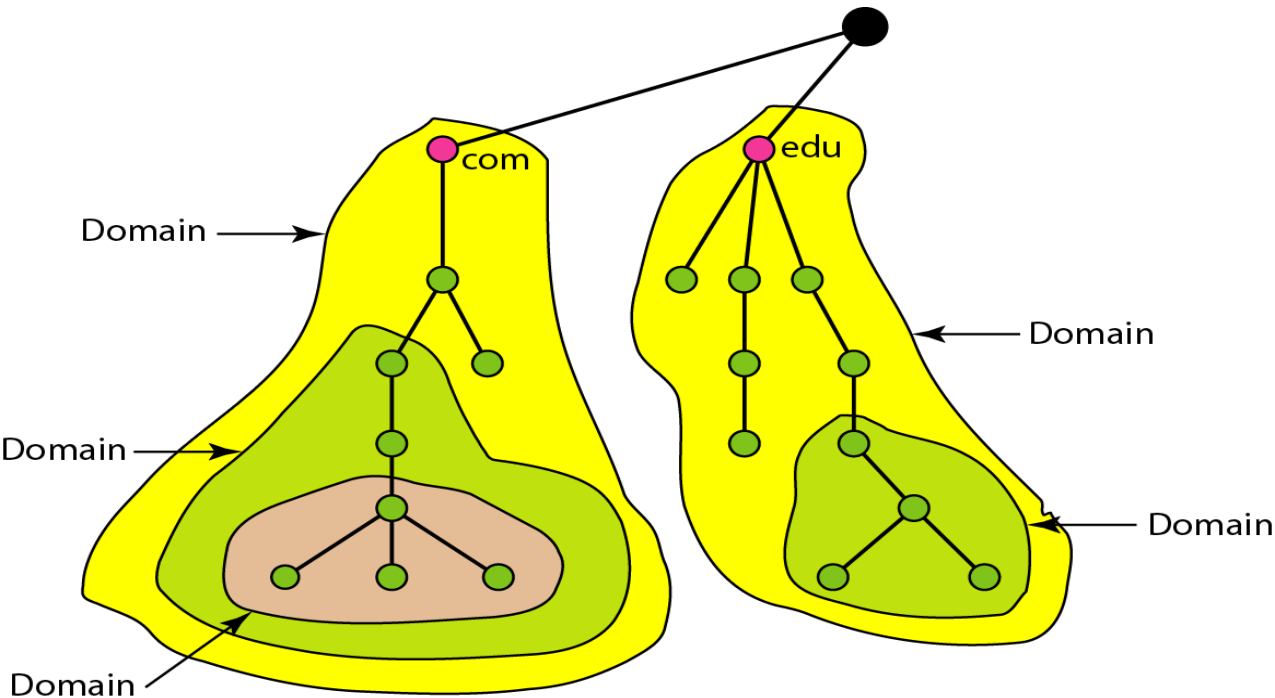
- **Client/server** network with a shared database
- **Domain** - Group of users, servers, and other resources
 - Share centralized account and security information in a database
- **Active Directory**
 - Contains domain database with objects, attributes and schema
 - Makes it easier to organize and manage resources and security

Active Directory -

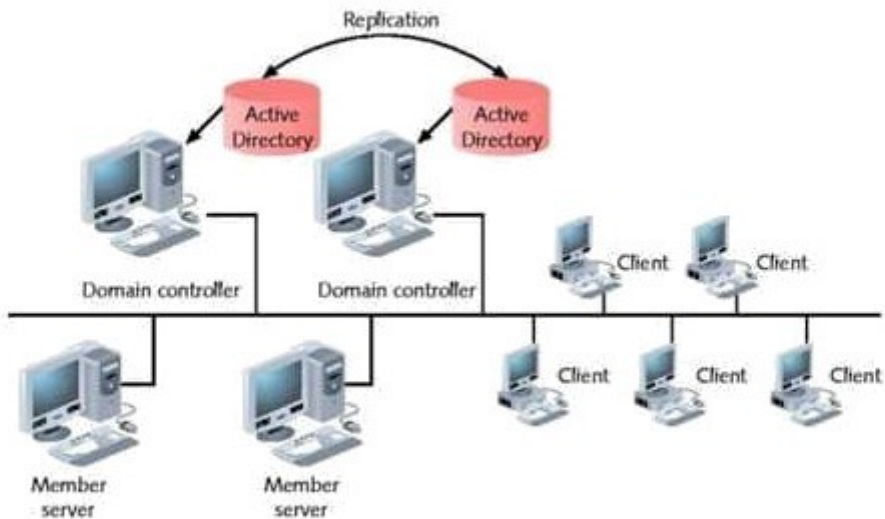
Domains

- **Domain** not confined by geographical boundaries
- **Domain controller servers**
 - **Contains directory information** about objects in a domain
- **Member servers**
 - Do not store directory information, can not be used to authenticate users
- **Replication**
 - Process of copying directory data to multiple domain controllers

Domains

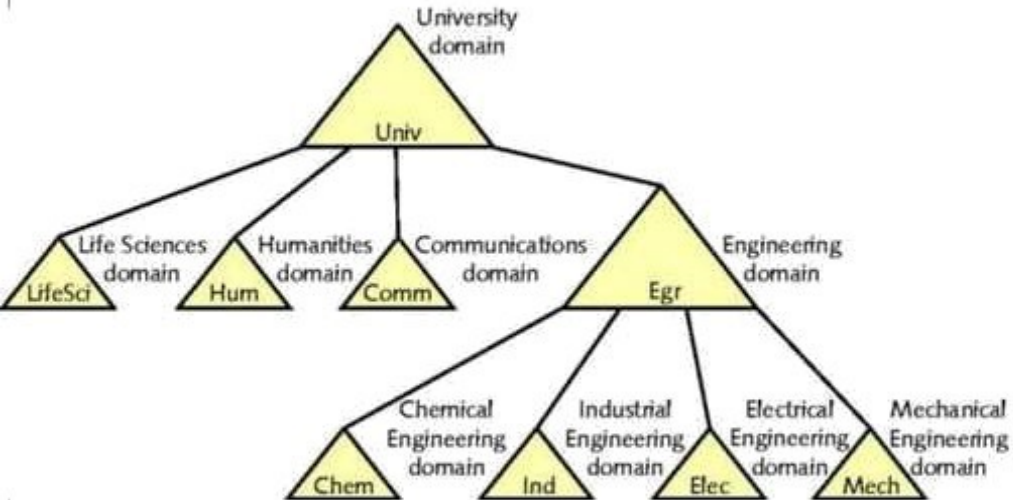


Domains



Domain model on a Windows Server 2008 network

Domains



Multiple domains in one organization

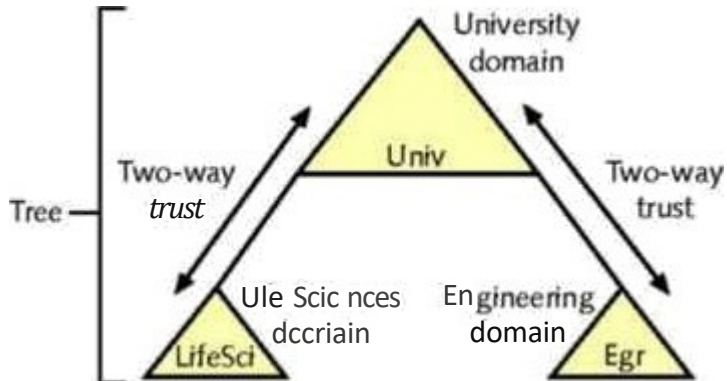
Trees

- Directory structure above domains
 - Large organizations use multiple domains
- **Domain tree**
 - Organizes multiple domains hierarchically
- **Root domain**
 - Active Directory **tree base**
- **Child domains**
 - Branch off from root domain

Trust Relationships

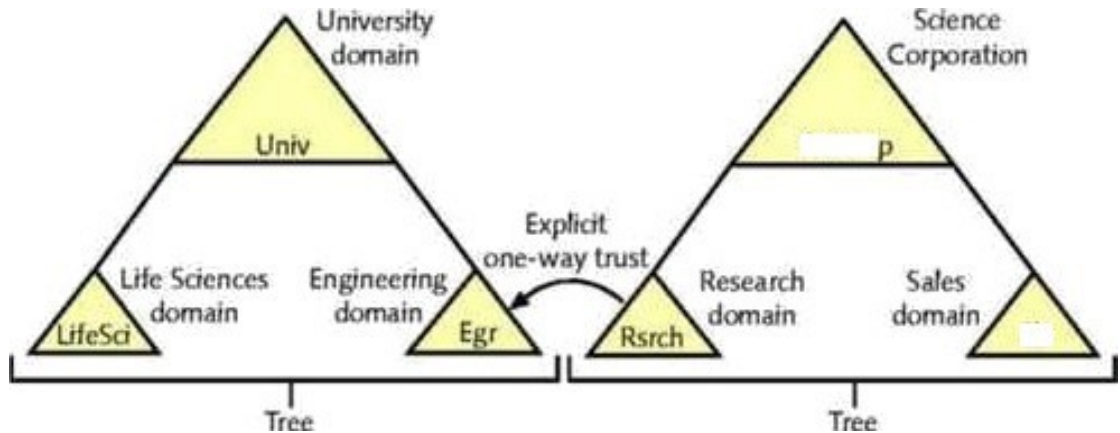
- Domains within the same tree
 - Share common Active Directory database
- Relationship between two domains
 - One domain allows another domain to authenticate its users
- Active Directory supports two trust relationship types
 - allows users to authenticate
 - **Two-way transitive trusts**
 - **Explicit one-way trusts**

Trust Relationships



Two-Way trusts between domains in a tree

Trust Relationships



Explicit one-way trust between domains in different trees

Namespaces



Namespaces

- Some namespaces are **flat**
 - there are **no duplicate** names
- Some namespaces are **hierarchical**
 - duplicate items within different branches of a tree
- Need **policies** to govern namespaces
 - Ideally, **written policies**
- Can become training for new System Admins (Sas)
- Needed to enforce adherence to policy

Namespaces Policies

- Naming policy
 - What names are permitted/not permitted?
- Technology — specific syntax
- Organizational — not offensive
- Standards compliance
 - How are names selected?
 - How are collisions resolved?
 - How do you merge namespaces?
- Technological and political concerns

Namespace policies (2)

Naming policy

- **How are names selected?**

1. **Formulaic**

- e.g., hostname: pc-0418: user-id: xyz210

2. **Thematic**

- e.g., using panel names for servers; coffee for printers

3. **Functional**

- e.g., specific-purpose accounts: admin, secretary, guest;
hostnames dns1, web3: disk partitions /finance. /dev

4. **Descriptive**

- e.g., location, object type (pl122-ps)

5. **No method**

- Everyone picks their pwn, first-come first-serve
- Once you choose one scheme, difficult to change — choose well!

Namespace policies (3)

Protection policy

What kind of protection does the namespace require?

- password list
- UIDs
- login IDs, e-mail addresses

Who can add/delete/change an entry?

- Need backups or change management to roll back a change

Namespace policies (4)

- **Scope Policy**

- Where is the namespace to be used?
 - How widely (geographically) shall it be used?
 - Global authentication is possible with RADIUS
 - NIS often provides a different space per cluster
 - How many services will use it? (thickness)
 - ID might serve for login, email, VPN, name on modem pools
 - Across different authentication services
 - Active Directory, NIS, RADIUS (even with different pw)
 - What happens when a user must span namespaces?
 - Different IDs? Confusing, lead to collisions
 - Single flat namespace is appealing; not always
-

Namespace policies (5)

- **Consistency policy**

- Where the same name is used in multiple namespaces, which attributes are also retained?

- E.g., UNIX name, requires same (real) person, same UID, but not same password for email, login

- **Reuse policy**

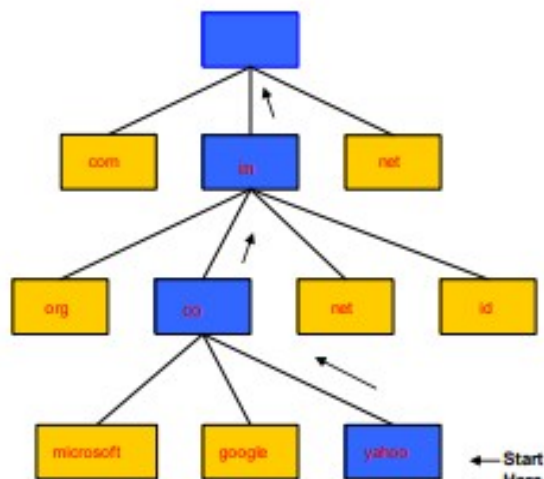
- How soon after deletion can the name be reused?

- Sometimes want immediate re-use (new printer)
- Sometimes long periods (prevent confusion and old email from being sent to new user)

DNS: The Domain Name System

- **What does DNS do?**
- The DNS namespace
- **How DNS works?**
- Testing and debugging (tools)

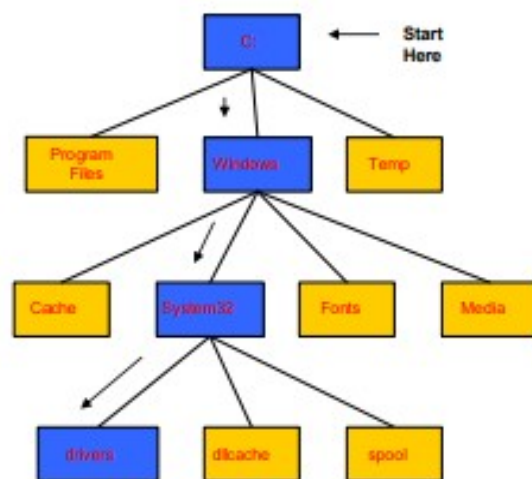
Naming a Domain



yahoo.co.in .

A "." is used as separator

Naming a Directory



C:\windows\system32\drivers\

A "\" is used as separator

What does DNS do?

- Provides **hostname** $\leftrightarrow$ **IP lookup services**
 - www.Iehigh.edu $\leftrightarrow$ 128.180.2.57
- DNS defines
 - A hierarchical namespace for hosts and IP addresses
 - A “resolver” — library routines that query this database
 - Improved routing for email
 - A mechanism for finding services on a network
 - A protocol for exchanging naming information
- DNS is essential for **any** org using the Internet

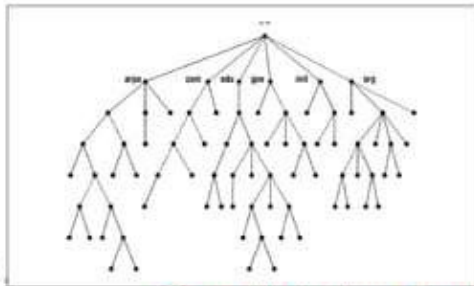
What uses DNS?

- Any application that operates over the Internet
- Such as
 - email
- Spam filters
 - WWW
- FTP
 - IRC,
 - Windows update
 - telnet, ssh



The DNS namespace

- A tree of “domains”
- Root is “.” (dot), followed by top-level (root-level) domains
- Two branches of tree
 - One maps hostnames to IP
 - Other maps IP address back to hostnames
- Two types of top-level domain names used today
 - **gTLDs**: generic top-level domains
 - **ccTLDs**: country code top-level domains



Some illustrations from
O'Reilly's DNS & Bind

Generic top-level domain names

Domain Purpose		Domain Purpose	
com	Companies	aero	Air transport industry
edu	Educational institutions	biz	Businesses
gov	(US) government agencies	coop	Cooperatives
mil	(US) military agencies	info	Unrestricted
net	Network providers	jobs	Human resources folks
org	Nonprofit organizations	museum	Museums
int	International organizations	name	Individuals
arpa	IP address lookup	pro	Professionals (attorneys, etc.)

But today there are an abundance of top-level domains

— .black, .blue, .airforce, .agency, .audio, etc.

- See <http://www.iana.org/domains/root/db/>

Common country code domain names

Code	Country	Code	Country
au	Australia	li u	Hungary
br	Brazil	jp	Japan
ca	Canada	md	Moldovia
et	Ethiopia	nsx	Mexico
Ch	Switzerland	nu	Niue
de	Germany	se	Sweden
fi	Finland	tm	Turkmenistan
fr	France	tv	Tuvalu
lik	Hong Kong	us	United States

- See <http://www.iana.org/domains/root/db/>

Domain Name Management

- Network Solutions (now **VeriSign**) used to manage .com, .org, .net, and .edu directly
- VeriSign now manages infrastructure for .com, .net, .tv, .name and .cc
 - Dozens of others manage country codes and other top-level domains
- Organizations can now register with many different registrars (even when VeriSign manages the underlying database)
- Domain holders **must have two name servers** authoritative for the domain

Selecting a Domain Name

- Most good (short) names in .com and other old gTLDs are already in use
- Domain names are up to 63 characters per segment (but a **12** character length limit is recommended), and up to 255 chars overall
- Identify two authoritative name servers
- Select a registrar, and pay -\$1-\$35/year for registration

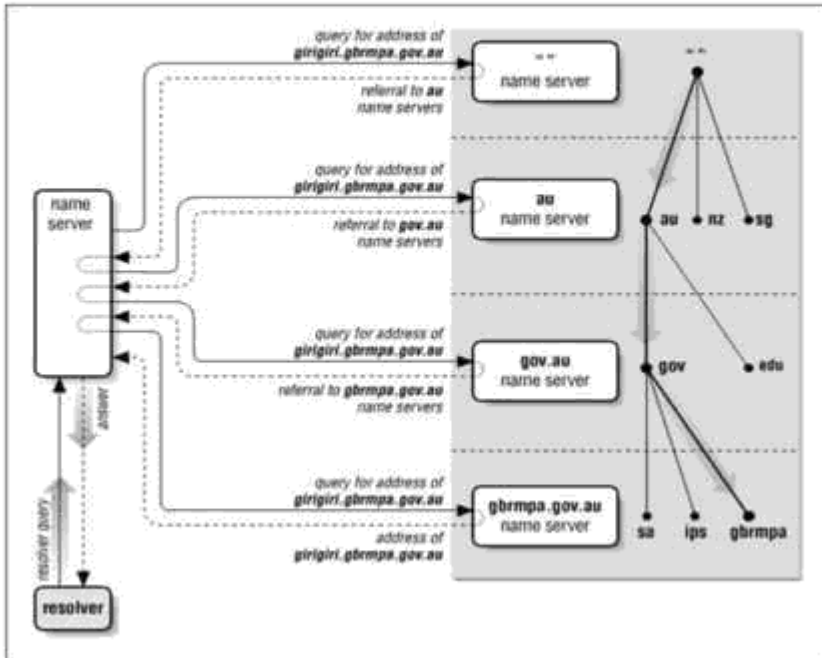
How DNS Works?

- A client calls **gethostbyname()**, which is part of the resolver library
- The resolver library sends a lookup request to the first nameserver that it knows about (from **/etc/resolv.conf**)
- If the nameserver **knows** the answer, it sends it back to the client
- If the nameserver **doesn't know**, it either
 - asks the next server, or
 - returns a failure, and suggests that the client contact the next server

What DNS Servers Know?

- All DNS servers know about the 13 root servers
 - hardcoded (rarely changes!), or in hint file
 - a.root-servers.net ... m.root-servers.net
- Each root DNS server knows about servers for every top-level domain (.com, .net, .uk, etc.)
- Each top-level domain DNS server knows the servers for each second-level domain server within the toplevel domain
- Authoritative servers know about their hosts

Example resolution



Types of name servers

- **Recursive** vs. **nonrecursive** servers

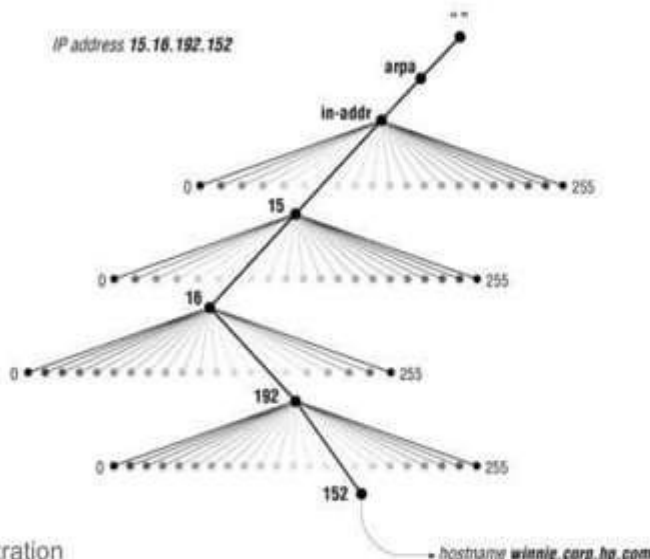
- Servers that allow recursive queries will do all the work
- Nonrecursive servers will only return referrals or answers

- **Authoritative** vs. **caching-only** servers

- Authoritative servers have the original data
- Caching servers retain data previously seen for

IP-to-hostname resolution

- IP resolution works essentially the same as hostname resolution
- Query for 15.16.192.152
 - Rendered as query for 152.192.16.15
 - 15.in-addr.arpa
- Each layer can delegate to the next



DNS on Linux

- Linux uses **/etc/nsswitch.conf** to determine what sources to use for name lookups

/etc/nsswitch.conf

passwd: files nisplus

shadow: files nisplus

group: files nisplus

hosts: files dns

- Configuration is in **/etc/named.conf**
- Other files in **/var/named**

Testing and debugging (tools)

- named supports lots of logging options
- typical BIND tools
 - nslookup (old, possibly deprecated)
 - whois - find domain and network registration info

Other Issues? (Reading Assignment)

- Other aspects of DNS haven't been covered in lecture
 - Lots of details!
 - Security issues
 - IPv6
 - Internationalization — now supported!
- DNS is generally case-insensitive
- VeriSign Site Finder product
 - See <http://cyber.law.harvard.edu/dns/>

Dynamic Host Configuration Protocol (DHCP)

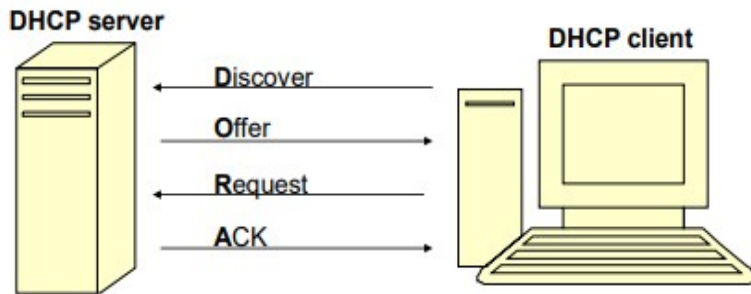
- Allows you to automatically assign IP addresses, subnet masks, and other configuration information to client computers on the local network.
- When a DHCP server is available, computers that are configured to obtain an IP address automatically request and receive their IP configuration from that DHCP server upon booting.
- When a DHCP server is unavailable, such clients automatically adopt an alternate configuration or an Automatic Private IP Addressing (APIPA) address

Benefits of DHCP

- Greatly **reduce the time** required to configure and
- Reconfigure ip addresses on your network.
- Allows you to **avoid configuration errors** resulting from entering IP address information manually at every host.
- Can **save the ip address** to be configured in the network

Benefits of DHCP CONT..

Process of obtaining TCP/IP Configuration



- **Discover:** The client will send a discover signal over the network to find a DHCP Server
- **Offer:** DHCP Server will give an offer after receiving the Discover signal from the Client
- **Request :** client will request DHCP server for an IP number after receiving an Offer from it.
- **Acknowledgment:** After receiving an request from the client the server will find free IP and sends a ACK signal.

Configuring DHCP Server

Implementing a basic DHCP server requires:

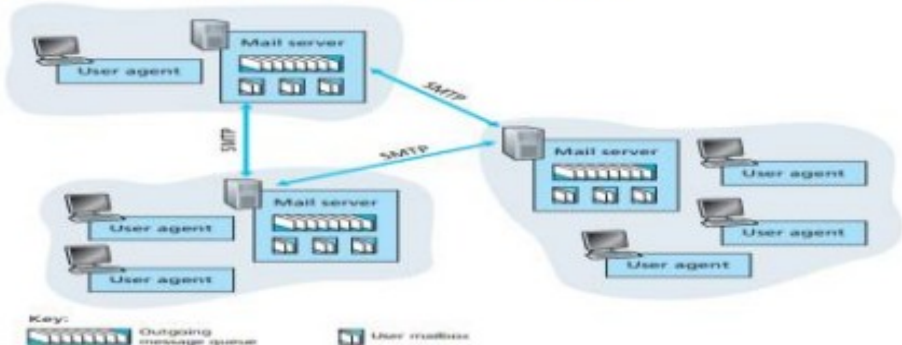
- installing the server
- **authorizing the server**
- configuring scopes, exclusions, reservations, and options
- **activating the scopes**
- finally, verifying the configuration

Web server

- It is Software services that support Web site creation, configuration, deployment, and management, along with other Internet functions.
- When a user requests a Web resource using **HTTP request**.
- The Web server responds with **HTTP response**.
- Such communication is done several times throughout a single session to a Web site.
- HTTP uses **TCP** for communication between clients and servers.
- HTTP operates on port **80**

E-mail server

- **Email** is an asynchronous communication
- **ROLE:** User Agent/Software, Mail Servers, SMTP (25)
- TCP based Protocol
- **User Software:** Microsoft Outlook and Thunderbird



E-mail Server CONT..

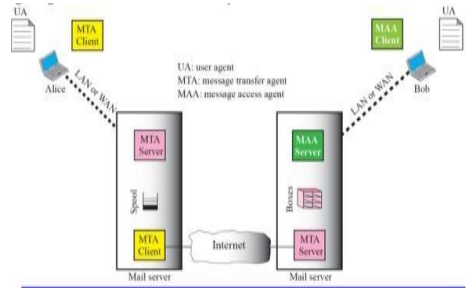
E-mail server

- **The three phases of transfer:**

- 1) handshaking (greeting)
- 2) transfer of messages
- 3) closure

- **Components of Email System**

- User Agent – PUSH
- Message Transfer Agent (SMTP) – PUSH
- Message Access Agents – PULL (POP3 or IMAP)



File Server

- local host User may want to transfer files to/from remote host.
- **Client:** Initiates transfer of files (to/from remote)
- **Server:** Remote computer for host files
- User must provide a **user identification** and a **password**
- User can transfer files from the local file system to the remote file system and vice versa.
- HTTP is used to view websites while **FTP** is used to access and transfer files.
- The common HTTP client is the browser while FTP can be accessed via the **command line** or a **graphical client** of its own

End of Chapter 2

Thank you !